

No.	Answer	Solution
12	C	Use $pV = nRT$, $T = \frac{pV}{nR}$ $= \frac{(1.0 \times 10^5)(10 \times 3.0 \times 4.0)}{5000 \times 8.31}$ $= 289 \text{ K}$ Mean translation KE of a gas molecule, $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$ $\langle c^2 \rangle = \frac{\frac{3}{2}(1.38 \times 10^{-23})(289)}{\frac{1}{2}\left(\frac{29 \times 10^{-3}}{6.02 \times 10^{23}}\right)}$ $c_{\text{r.m.s.}} = 498$ $\approx 500 \text{ m s}^{-1}$
13	D	Amount of thermal energy supplied to the water